Chapter 13 worked solutions

Structured State Spaces and Long Convolutions

This guide covers Exercises 1–4 and 10–11. It follows a state from one input to the next, reads the same calculation as a convolution, and compares the history that different updates need to carry. The companion Jupyter notebook works through Exercises 5–9 and is also available as a PDF.

1. Follow one state

Start with an empty memory, represented by the number $x_0 = 0$. Every step keeps one quarter of the old state and then adds the new input in full. Position $t = 1$ reads the first input, so x_1 is the state **after** reading u_1. This distinction matters when we count how often the first input is multiplied by 1/4.

Position t	New input u_t	Retained part $\frac{1}{4}x_{t-1}$	State after adding the input
1	4	$\frac{1}{4}(0) = 0$	$x_1 = 0 + 4 = 4$
2	0	$\frac{1}{4}(4) = 1$	$x_2 = 1 + 0 = 1$
3	0	$\frac{1}{4}(1) = \frac{1}{4}$	$x_3 = \frac{1}{4} + 0 = \frac{1}{4}$
4	2	$\frac{1}{4}(\frac{1}{4}) = \frac{1}{16}$	$x_4 = \frac{1}{16} + 2 = \frac{33}{16}$

The states are $(4, 1, \frac{1}{4}, \frac{33}{16})$, or $(4, 1, 0.25, 2.0625)$. At position 4, the first input has been carried across three transitions, from position 1 to 2, 2 to 3, and 3 to 4. Its contribution is $4(1/4)^3 = 1/16$. The new input contributes 2 immediately, with no retention factor yet. Inputs 2 and 3 are zero, so those two contributions account for the whole final state.

2. Read by lag

We can work out the same states by asking how much of each input reaches a particular output. If an input arrived at the current position, its **lag** is zero. The input one step back has lag one. In general, input u_s has lag $\ell = t - s$ when we read y_t. Here the output is the state itself, so there is no additional multiplication when we read it.

An input enters with coefficient 1, and each transition multiplies its remaining contribution by 1/4. Therefore the filter tap at lag ℓ is $k_\ell = (1/4)^\ell$. The first four taps, from the newest input to the oldest, are

$$(k_0, k_1, k_2, k_3) = \left(1, \frac{1}{4}, \frac{1}{16}, \frac{1}{64}\right).$$

Use those taps with the inputs $(u_1, u_2, u_3, u_4) = (4, 0, 0, 2)$. The first tap always multiplies the input at the position whose output we want. Each successive tap moves one position farther back. For y_3 only three input positions exist, so the tap for lag three does not enter the sum. For y_4 all four taps apply.

$$
\begin{aligned}
y_3 &= k_0u_3 + k_1u_2 + k_2u_1 \\
&= 1(0) + \tfrac{1}{4}(0) + \tfrac{1}{16}(4) = \tfrac{1}{4}, \\
y_4 &= k_0u_4 + k_1u_3 + k_2u_2 + k_3u_1 \\
&= 1(2) + \tfrac{1}{4}(0) + \tfrac{1}{16}(0) + \tfrac{1}{64}(4) = \tfrac{33}{16}.
\end{aligned}
$$

These are the same values as x_3 and x_4 in Exercise 1. The recurrence applies the retention factors as it moves forward, while the convolution uses the accumulated factor for each lag directly. For instance, the first input needs two multiplications by $1/4$ to reach y_3, and the filter combines both multiplications into $k_2 = 1/16$.

3. Carry two features

Now the state has two coordinates. The first retains half of its previous value at each step. The second retains three quarters. Each new input is added to both coordinates because $\bar{B} = (1, 1)^\top$. The read vector $C = (1, -1)$ then subtracts the second coordinate from the first.

Write the state as $\mathbf{x}_t = (p_t, q_t)^\top$, where p_t is the first coordinate and q_t is the second. The diagonal transition $\bar{A} = \operatorname{diag}(1/2, 3/4)$ gives

$$
p_t = \tfrac{1}{2}p_{t-1} + u_t, \qquad q_t = \tfrac{3}{4}q_{t-1} + u_t, \qquad y_t = p_t - q_t.
$$

Start from $(p_0, q_0) = (0, 0)$ and use $u = (4, 0, 0)$.

Position t	Input	First coordinate p_t	Second coordinate q_t	State $\mathbf{x}_t$	Output y_t
1	4	$\frac{1}{2}(0) + 4 = 4$	$\frac{3}{4}(0) + 4 = 4$	$(4, 4)^\top$	$4 - 4 = 0$
2	0	$\frac{1}{2}(4) + 0 = 2$	$\frac{3}{4}(4) + 0 = 3$	$(2, 3)^\top$	$2 - 3 = -1$
3	0	$\frac{1}{2}(2) + 0 = 1$	$\frac{3}{4}(3) + 0 = \frac{9}{4}$	$(1, \frac{9}{4})^\top$	$1 - \frac{9}{4} = -\frac{5}{4}$

The outputs are $(0, -1, -\frac{5}{4})$. The first output is zero because the two coordinates are equal, even though both store a nonzero value. After the input becomes zero, the transition still changes the stored state. The first coordinate fades faster, so the balance between them changes and the readout becomes negative. With no new input, the change comes entirely from the stored values.

4. Choose a step size

With $A = -1$, the continuous system tends to shrink a nonzero state when there is no input. The bilinear rule converts this continuous behavior into an update once per position. It produces a retention coefficient $\bar{A}$ for the old state and a write coefficient $\bar{B}$ for the new input. The scalar identity is 1, so the inverse factor in both formulas is division by $1 - \Delta A/2$.

Substitute $A = -1$ and $B = 1$ before choosing the step size.

$$
\bar{A} = \frac{1 + \Delta A/2}{1 - \Delta A/2} = \frac{1 - \Delta/2}{1 + \Delta/2}, \qquad \bar{B} = \frac{\Delta B}{1 - \Delta A/2} = \frac{\Delta}{1 + \Delta/2}.
$$

Step size Δ	Shared denominator $1 + \Delta/2$	Retention $\bar{A}$	Input write $\bar{B}$
$\frac{1}{2}$	$\frac{5}{4}$	$(\frac{3}{4})/(\frac{5}{4}) = \frac{3}{5}$	$(\frac{1}{2})/(\frac{5}{4}) = \frac{2}{5}$
1	$\frac{3}{2}$	$(\frac{1}{2})/(\frac{3}{2}) = \frac{1}{3}$	$1/(\frac{3}{2}) = \frac{2}{3}$

When the input is zero, its write term $\bar{B}u_t$ vanishes. The new state is just $\bar{A}x_{t-1}$. In one update, $\Delta = 1/2$ **retains more** of the magnitude of a nonzero state because $3/5 > 1/3$. For example, an old state of 15 becomes 9 at step size 1/2 and 5 at step size 1. Each number describes one discrete update with the indicated step size.

10. Count streaming storage (optional)

Imagine the next input has yet to arrive. Count the numerical history the layer has to carry across that boundary, per feature. A state space layer with 16 state coordinates stores the current vector of 16 numbers. When the next input arrives, the layer uses that vector to form the new state. The count is 16 regardless of how many steps have already passed.

A direct streaming convolution with a length-L filter forms the next output from the incoming sample and the $L - 1$ preceding samples. The incoming sample supplies lag zero and is available at the new step. Once the history is filled, it contains the samples for lags 1 through $L - 1$. At startup, positions before the first input contribute zero. For example, a three-tap filter uses the current input and two older inputs. After producing the output it keeps the two most recent samples for the next step.

Filter length in the direct method	State space layer, carried numbers	Direct convolution, carried numbers
1,024	16	$1,024 - 1 = 1,023$
2,048	16	$2,048 - 1 = 2,047$

Doubling the filter length adds 1,024 retained samples to the direct method and leaves the state width at 16. The counts concern feature values carried between steps. The filter taps and the transition weights are model parameters, and temporary products and sums are working storage. Neither enters the table. The direct method preserves recent inputs individually, while the state vector stores the result of its update rule.

11. Check a changing rule (optional)

In this exercise the retention factor can change from step to step. Because $x_0 = 0$, the first state is $x_1 = u_1$. We can then read the effect of each input by substituting the earlier state into the update.

$$\begin{aligned} y_2 = x_2 &= a_2 x_1 + u_2 = \tfrac{1}{2} u_1 + u_2, \\ y_3 = x_3 &= a_3 x_2 + u_3 = \tfrac{3}{4}\left(\tfrac{1}{2} u_1 + u_2\right) + u_3 \\ &= \tfrac{3}{8} u_1 + \tfrac{3}{4} u_2 + u_3. \end{aligned}$$

The coefficient of u_1 in y_2 is $1/2$. The coefficient of u_2 in y_3 is $3/4$. In both comparisons, the output comes exactly one position after the input. The difference comes from the transition across that gap, which uses a_2 in the first comparison and a_3 in the second.

A fixed causal filter uses one lag-one coefficient k_1 at every position. To see the conflict, put a 1 at position 1 with all other inputs zero. One step later the recurrence gives $1/2$. Put the 1 at position 2 instead, again with all other inputs zero. One step later the recurrence gives $3/4$. The same fixed tap cannot reproduce both responses. When the retention factors are prescribed but vary with position, the update is still linear in the inputs, but the weight of an input depends on which transitions it crosses. A lag alone no longer identifies that weight.